Week 2 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- **M4:** perform matrix operations; compute simple determinants and inverses; use Gaussian elimination; compute rank; solve a linear system; and classify a system as having a unique solution, no solution or multiple solutions; and
- **S2:** compute and interpret conditional probabilities; use or test independence; and apply the law of total probability and Bayes' rule.

Prerequisites and retrieval

You will use algebra and functions from `M2` and event/probability rules from `S1` throughout the new material. The final question contains three explicit Week 1 retrieval parts on logarithms (`M2`), Taylor approximation (`M3`) and set notation with elementary counting/probability (`M1`, `S1`).

Before starting, check that you can rearrange a linear equation, multiply signed numbers and fractions, use set notation for events, and apply the addition rule for two events.

Student materials

1. Read the exact external sections and skip instructions in the reading guide.
2. Study and reproduce the original worked examples.
3. Attempt every part in the exercise sheet without consulting the full solutions.
4. Use the progressive hints only after a serious attempt.
5. Mark your work with the complete solutions, record errors by outcome ID, and reattempt missed parts after two or three days.
6. Finish with the 25-minute self-test, then use its solutions to update your error log.

The required external readings are Jim Hefferon's *Linear Algebra*, at the listed chapter and subsection locators, and MIT OpenCourseWare 18.05 Class 3 Reading, §§2–7.2. Do not read or attempt the excluded software, video or out-of-scope sections named in `reading_guide.md`.

Suggested study sequence

1. Review matrix dimensions and multiplication.
2. Work through Gaussian elimination, pivots and rank.
3. Connect rank to the three possible classifications of a linear system.
4. Review conditional probability, then derive total probability and Bayes' rule from the multiplication rule.
5. Alternate linear-algebra and probability questions rather than completing one entire strand first.
6. Finish with the Week 1 retrieval block, mark all work, and update your error log.

Time plan

The core workload is about 9 hours 20 minutes.

The times below are approximate planning estimates and have not yet been verified by students. Students in the first live course run will be invited to report their actual component times so the plan can be recalibrated for the following year.

Activity	Suggested time
Required external reading	155 minutes
Institutional notes and worked-example reproduction	105 minutes

Activity	Suggested time
Exercise sheet	240 minutes
Self-check, marking, error logging and first reattempt	60 minutes
Total	**560 minutes**

Required outputs

By the end of the week, you should have:

- handwritten or typed notes stating the rank solvability criterion and the four central probability formulas;
- your own reproduction of the worked elimination and Bayes examples;
- a complete first attempt at all 17 scorable parts;
- marked corrections for each error, labelled `M4`, `S2` or the relevant Week 1 retrieval outcome; and
- a dated plan to reattempt every missed part after two or three days.

There is no optional or non-examined material in this Week 2 package.

Required reading route

Complete these two required written selections before using the institutional worked examples. The exact locators and skip instructions are part of the assignment: do not treat either whole source as required. No video, applet, code or specialist software is required.

Linear algebra (`M4`)

Read Jim Hefferon, *Linear Algebra*, at these locators:

- Chapter One I.1, “Gauss’s Method”;
- Chapter One I.2, “Describing the Solution Set”;
- Chapter One III.1, “Gauss–Jordan Reduction”;
- Chapter Three IV.1–IV.4, “Matrix Operations”; and
- Chapter Four I.2, “Properties of Determinants”.

Skip: chapter-end Topics, especially Computer Algebra Systems; the Sage lab and videos; determinant permutation proofs; Cramer’s Rule; and every software computation.

Course resource record: `MATH-M4-HEF-REQ`; source record: `HEFFERON_LA`.

Target reading time: 100 minutes.

Conditional probability (`S2`)

Read Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05, Class 3 Reading, §§2–7.2.

Skip: Problem Set 2 Problem 6, which requires R, and all linked videos, interactive questions and applets. Do not treat the whole course as required.

Course resource record: `STAT-S2-REQ-MIT`; source record: `MIT_1805`.

Target reading time: 55 minutes.

Reading output

After each selection, close the source and write a brief original summary. For `M4`, state the rank test for unique, multiple and no-solution cases. For `S2`, state conditional probability, independence, total probability and Bayes’ rule with their conditions. Do not copy external explanations or exercise prompts into shared notes.

Worked examples

These original institutional notes and examples support `M4` and `S2`. Work through each **Method** before reading its **Solution**, then reproduce the calculation from a blank page. The examples do not reproduce an external exercise or solution.

The notes and examples below are original course material. All vectors are column vectors.

Matrices and their operations

An $m \times n$ matrix has m rows and n columns. Two matrices can be added only when their dimensions are the same. If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ exists and is $m \times p$. Its entry in row i and column j is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

The order matters. Even when both products exist, $\mathbf{AB}$ and $\mathbf{BA}$ need not be equal and may have different dimensions.

The transpose $\mathbf{A}^{\mathsf{T}}$ interchanges rows and columns. Thus, if $\mathbf{A}$ is $m \times n$, then $\mathbf{A}^{\mathsf{T}}$ is $n \times m$.

Example: checking dimensions before multiplying Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Method. The matrix is 2×3 and the vector is 3×1, so $\mathbf{Ab}$ exists and is 2×1.

Solution. Row-by-column multiplication gives

$$\mathbf{Ab} = \begin{bmatrix} 1(2) + (-1)(1) + 2(-1) \\ 0(2) + 3(1) + 1(-1) \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Interpretation. A dimension check prevents an undefined product and also predicts the dimension of the answer.

Linear systems and Gaussian elimination

A system of m linear equations in n unknowns can be written

$$\mathbf{Ax} = \mathbf{b},$$

where $\mathbf{A}$ is $m \times n$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. The augmented matrix is $[\mathbf{A} \mid \mathbf{b}]$.

The elementary row operations are:

1. interchange two rows;
2. multiply a row by a non-zero scalar; and
3. add a scalar multiple of one row to another row.

These operations preserve the solution set. Gaussian elimination uses them to reach row-echelon form. A leading non-zero entry in a non-zero row is a pivot. Back-substitution then recovers the unknowns. Gauss–Jordan elimination continues until each pivot is the only non-zero entry in its column.

Example: a system with a free variable Consider

$$\begin{aligned} x + 2y - z &= 3, \\ 2x + y + z &= 4, \\ 3x + 3y &= 7. \end{aligned}$$

Method. Form the augmented matrix and eliminate x from rows 2 and 3.

Solution. Begin with

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 1 & 1 & 4 \\ 3 & 3 & 0 & 7 \end{array}\right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 3 & -2 \\ 0 & -3 & 3 & -2 \end{array}\right].$$

Then $R_3 \leftarrow R_3 - R_2$, so the third row is zero. There are only two pivots, not three. Let $z = t \in \mathbb{R}$. The second row gives $-3y + 3t = -2$, so $y = t + 2/3$. The first row gives $x = 5/3 - t$. Hence

$$\mathbf{x} = \begin{bmatrix} 5/3 - t \\ 2/3 + t \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Interpretation. An apparently square system can still have multiple solutions. The number of equations alone does not determine solvability.

Rank and solvability

The rank of a matrix is the number of pivots in any row-echelon form. Write it as $\operatorname{rank}(\mathbf{A})$. For a system with n unknowns:

- if $\operatorname{rank}(\mathbf{A}) \neq \operatorname{rank}([\mathbf{A} \mid \mathbf{b}])$, the system has no solution;
- if the two ranks are equal to n, the system has a unique solution; and
- if the two ranks are equal but are less than n, the system has multiple solutions, with at least one free variable.

A row of the form

$$\begin{bmatrix} 0 & \cdots & 0 \mid c \end{bmatrix}, \qquad c \neq 0,$$

is a contradiction. It increases the augmented rank but not the coefficient rank, so the system is inconsistent.

Example: a parameter changes consistency Suppose elimination gives

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & \theta - 4 \end{array}\right].$$

If $\theta \neq 4$, the last row is a contradiction, so the system has no solution. If $\theta = 4$, the last row is zero. There are two pivots and three unknowns, so the consistent system has multiple solutions.

Determinants and inverses

For a 2×2 matrix,

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \det(\mathbf{A}) = ad - bc.$$

The matrix is invertible exactly when $\det(\mathbf{A}) \neq 0$. In that case,

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

When $\mathbf{A}$ is an invertible $n \times n$ matrix, the system $\mathbf{Ax} = \mathbf{b}$ has the unique solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. Elimination remains the more general method because it also diagnoses singular and non-square systems.

Conditional probability

For events A and B with $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Rearranging gives the multiplication rule,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Conditioning changes the relevant reference group. The denominator is the probability of the event after the vertical bar.

Independence

Events A and B are independent when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If both conditioning events have positive probability, equivalent statements are $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ and $\mathbb{P}(B \mid A) = \mathbb{P}(B)$. Independence is not the same as disjointness. Two disjoint events with positive probability cannot be independent because their intersection has probability zero while the product of their probabilities is positive.

Total probability and Bayes' rule

Let $B_1, \ldots, B_k$ be a partition of Ω: the events are pairwise disjoint, their union is Ω, and each has positive probability. The law of total probability states that

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

For any j with $\mathbb{P}(A) > 0$, Bayes' rule gives

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

The numerator is the joint probability of A and B_j. The denominator adds the joint probabilities for every route by which A can occur.

Example: total probability and reversal of conditioning A department offers a morning session to 3/5 of students and an evening session to 2/5. Let M and E denote the two session events, and let C denote completing a diagnostic correctly. Suppose

$$\mathbb{P}(C \mid M) = \frac{2}{3}, \qquad \mathbb{P}(C \mid E) = \frac{1}{2}.$$

Method. The events M and E form a partition. First find $\mathbb{P}(C)$, then reverse the conditioning with Bayes' rule.

Solution. Total probability gives

$$\mathbb{P}(C) = \frac{2}{3}\frac{3}{5} + \frac{1}{2}\frac{2}{5} = \frac{3}{5}.$$

Therefore,

$$\mathbb{P}(M \mid C) = \frac{\mathbb{P}(C \mid M)\mathbb{P}(M)}{\mathbb{P}(C)} = \frac{(2/3)(3/5)}{3/5} = \frac{2}{3}.$$

Interpretation. The conditional probability must be computed in the direction asked. In general, $\mathbb{P}(M \mid C)$ is not the same as $\mathbb{P}(C \mid M)$, even though they happen to be numerically equal in this example.

Common pitfalls

- Check matrix dimensions before multiplying.
- Record row operations so an arithmetic error can be located.
- Do not infer uniqueness merely because the system is square.
- Compare the coefficient and augmented ranks before classifying a system.
- Read $\mathbb{P}(A \mid B)$ as "probability of A, given B".
- Do not interchange $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$.
- Test independence with a valid equality; do not infer it from a verbal impression.
- In total probability, verify that the conditioning events form a partition.